INTEROFFICE MEMORANDUM

To: CIO

From: Team 02

Date: October 24, 2026

Subject: Role of AI in Cyber Defense

Summary: This memo discusses how AI technology can be useful in providing Cyber Defense capabilities to an organization.

We hope you find this report helpful for use in the next board of directors meeting.

Artificial Intelligence, almost synonymous with machine learning, has revolutionized the Cybersecurity industry. From AI enable next-generation firewalls for intrusion detection and prevention systems, to smart filtering and blocking of phishing emails, as McKinsey & Company puts it, “AI is the greatest threat-and defense-in Cybersecurity today. For the sake of our organization, it’s been our greatest asset.

Over the past few hours, while combating active incidents, our security team has been able to both thwart attacks and troubleshoot critical business services leveraging free AI platforms. This has substantially decrease our time-to-resolve incidents, and troubleshooting critical service failures.

For example, troubleshooting issues with our SIEM monitoring tools. AI was used to help us understand why our Webmail Roundtable service was inaccessible as it has a strong ability to parse large amounts of data and pinpoint configuration errors.

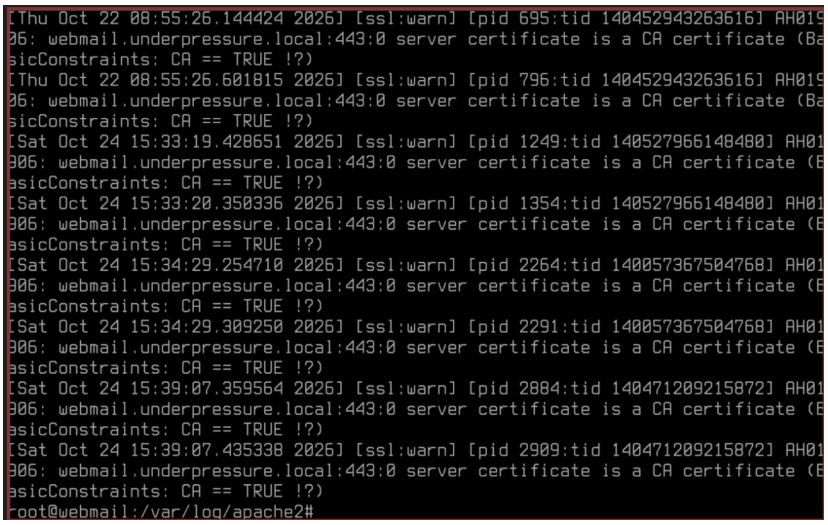


Figure 1: Dense logs from our Roundtable Web server.

AI has also been incredibly impact for our security engineers when it comes to the creation of custom tooling, monitoring, and further security control development. For example, we leveraged AI to create scripts to automatically install ClamAV on our Linux devices which software now scans our devices to identify malware.

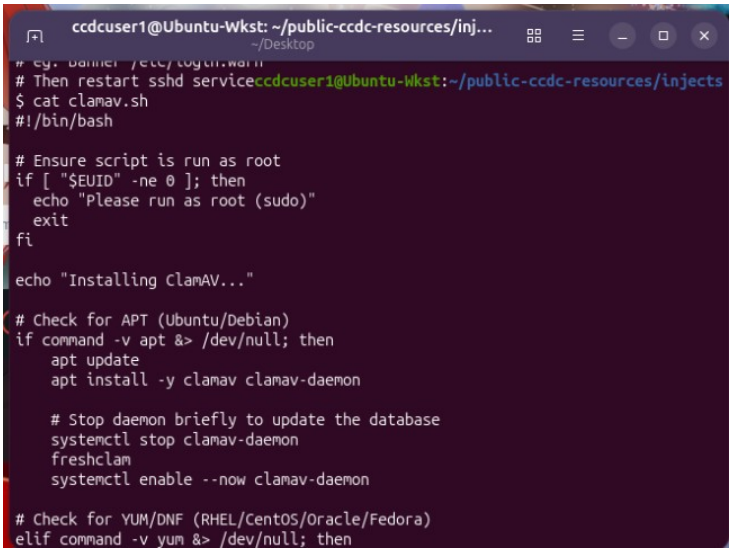


Figure 2: ClamAV script that was utilized on our devices.

By leveraging AI for this service level troubleshooting, we were able to determine the root cause of the service downtime and remediate almost immediately.

Moving forward, we believe that integrating AI agents, services, and systems will be how we differentiate our security posture from other organizations. One goal that our team has moving forward is the creation of a threat hunting, incident response AI enabled agent to continuously monitor our networks, identify indicators of compromise, and further protect our environment.

Effective use of AI in our organization could potential lead to the reduction of the number of team members required to monitor and secure our network. However, Artificial Intelligence can make mistakes and needs knowledgeable humans to review and correct its work and report. AI is an extremely effective tool in identify malicious activity, troubleshooting, and implementing security solutions into our systems but Humans will need to be involved in every step of the process.

However, with the small size of our business network, investing a large amount of money in AI tools may be unnecessary as Humans can easily manage the current load of configuration needs and log analysis with free-tier models that we have access to.

We are firm believers that integrating AI into our Cybersecurity applications, tooling, services, and workflows, in a responsible and practical fashion, will allow us to be at the forefront of technology, adding security layers throughout our organization, and optimizing for overall availability for our customers by establishing advanced security controls.

If there are any concerns, questions, or clarifications, please do not hesitate to reach out.

With Regards,

Team 02

Sources:

https://www.mckinsey.com/about-us/new-at-mckinsey-blog/ai-is-the-greatest-threat-and-defense-in-cybersecurity-today